CONSTRUCTOR

- A class ***constructor*** is a special method called only once per object, at the time of its creation.
- The constructor has the same name as the class constructor.
- The role of a constructor is to initialize the attributes of an object (initialization method).
- The constructor is generally public and does not have a return type (not even "void").
- A class can have several constructors that differ in the number and nature of their parameters.
- In the absence of a constructor defined by the programmer, Java provides a constructor with no parameters that initializes the attributes of the created object (0 for numeric values, Null for references, and False for Booleans).

CONSTRUCTOR

- Example:

```
public class Application
{
public static void main (String args[])
    {
        Person Jean = new Person ();
        Jean.setName( "jean");
    }
}
```

The constructor used here is the default one (no constructor is defined in the Person class).

CONSTRUCTOR

Person.java

```
public class Person {

  public String Name, LastName;
  public int age;

  public Person (String aName, String
aLastName, int anAge)
  {
     Name= aName;
     LastName= aLastName;
     age = anAge;
  }
}
```

Defining a constructor. The default constructor (Person()) no longer exists. The preceding code generates an error.

To access the attributes or methods of an object, we use the dot "."

```
public class Application
{
  public static void main (String args [])
     {
        Person Jean=new Person();
        Jean.setName( "jean");
     }
}
```

This will throw a compilation error.

CONSTRUCTOR

- Example constructors

1. Public Person (int x)
{
  age = x;
}
2. Public Person (String S)
{
  Name= S;
}
3. Public Person (String S, int x)
{
  Name= S;
  age = x;
}

- Example of objects

1. Person p1 = new Person (10); → constructor 1.
2. Person p2 = new Person ("Ali", 10); → constructor 3.
3. Person p3 = new Person("Ali"); → constructor 2.
4. ~~Person p4 = new Person (); → error~~

CONSTRUCTOR

```
class Circle()
{
   public double x,y; //Center coordinates
   private double r; //Circle radius
   public Circle (double r1) {
      r = r1;
   }
    public double area (){
    return 3.14159*r*r;
   }
}
```

```
public class MyProgram()
{
   public static void main (String [] args){
      Circle c; // c is a reference to a Circle object, not an object
      c = new Circle (5);/* c now references an object allocated in memory*/
     c.x=c.y=10;
     System.out.println(" Area of c: " +c.area());
   }
}
```


EXERCISE: THE CIRCLE CLASS

Instructions:

- **Create a class** named Circle.
- **Define one attribute**: radius (double).
- **Implement 4 methods**:
  - setRadius(double r): To assign a value to the radius.
  - calculateCircumference(): Returns the distance around the circle (2*pi*radius).
  - calculateArea(): Returns the surface area (pi *radius^2).
  - displayInfo(): Prints the radius, circumference, and area.
- Create a class named TestCircle witch contain main method, create a circle object, shows the initial state of the circle, modify the radius and shows the new results.

EXERCISE: THE RECTANGLE CLASS

- Write a Rectangle class to display Rectangle objects.
- Each Rectangle object has the attributes width and length of type float and the methods:
  - perimeter(): which calculates and returns the perimeter of the rectangle.
  - diagonal(): which calculates and returns the value of the diagonal of the rectangle.
  - double(): which doubles the length of the rectangle.
- Write a TestRectangle class that contains a main method calling the methods perimeter(), diagonal(), and double() and displays their results.

EXERCISE: THE BOOK CLASS

1. Create a class named Book.
2. Add three attributes: title (String), author (String), and price (double).
3. Create a **constructor** that initializes these three attributes.
4. Create a method named displayDetails() that prints the book's info in this format:

"Title: [title], Author: [author], Price: [price]"

5. In your main method, create two different books and call displayDetails() for each.